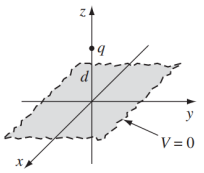
H3: Potentials

3.2 the method of images

charge above a conductor

Consider a point charge q held a distance d above an infinite grounded conducting plate
> the potential above the plate is determined by q, but also the induced charge on the plate
> this is a hard problem to solve, but we have the boundary conditions:

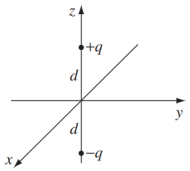
1. $V = 0$ when $z = 0$ (since the conducting plane is grounded).
2. $V \to 0$ far from the charge (that is, for $x^2 + y^2 + z^2 \gg d^2$).



instead, consider two point charges q at (0,0,d) and -q at (0,0,-d)
> the potential is:

$$V(x, y, z) = \frac{1}{4\pi\epsilon_0}\left[\frac{q}{\sqrt{x^2 + y^2 + (z - d)^2}} - \frac{q}{\sqrt{x^2 + y^2 + (z + d)^2}}\right]$$

it follows that
1. $V = 0$ when $z = 0$,
2. $V \to 0$ for $x^2 + y^2 + z^2 \gg d^2$,



> these are the exact same conditions as our original problem
> the second problem produces the exact same potential in region z>0
> we can use the equation to solve our first problem

3.2.2 induced surface charge

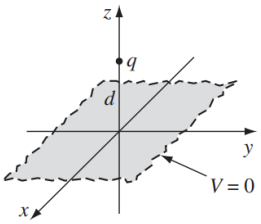
induced surface charge

we can compute the surface charge via:

$$\sigma = -\epsilon_0 \frac{\partial V}{\partial n}$$

with ∂V/∂n the normal derivative of V at the surface
> this is in the z-direction for this problem:

$$\sigma = -\epsilon_0 \left.\frac{\partial V}{\partial z}\right|_{z=0}$$



we can put this in the known eq for the potential (see above)

$$\frac{\partial V}{\partial z} = \frac{1}{4\pi\epsilon_0}\left\{\frac{-q(z - d)}{[x^2 + y^2 + (z - d)^2]^{3/2}} + \frac{q(z + d)}{[x^2 + y^2 + (z + d)^2]^{3/2}}\right\}$$

thus:

$$\sigma(x, y) = \frac{-qd}{2\pi(x^2 + y^2 + d^2)^{3/2}}$$

which is a negative charge greatest at x=y=0

total induced charge

we can calculate the total induced charge Q via:

$$Q = \int \sigma\, da.$$

which we will do in polar coords (r,ϕ) with $r^2=x^2+y^2$ and da=rdrdϕ:

$$\sigma(r) = \frac{-qd}{2\pi(r^2 + d^2)^{3/2}}$$

and:

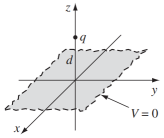
$$Q = \int_0^{2\pi}\int_0^{\infty} \frac{-qd}{2\pi(r^2 + d^2)^{3/2}} r\, dr\, d\phi = \left.\frac{qd}{\sqrt{r^2 + d^2}}\right|_0^{\infty} = -q$$

3.2.3 force and energy

force on the charge

the charge q is attracted to the plane due to the induced charge
> we can use the second example, since the fields are the same
> the force is also the same:

$$\mathbf{F} = -\frac{1}{4\pi\epsilon_0}\frac{q^2}{(2d)^2}\hat{\mathbf{z}}.$$



energy of the charge

calculate the work needed to bring the charge at a distance d from infinity:

$$W = \int_{\infty}^{d} \mathbf{F} \cdot d\mathbf{l} = \frac{1}{4\pi\epsilon_0}\int_{\infty}^{d} \frac{q^2}{4z^2}\, dz$$

$$= \frac{1}{4\pi\epsilon_0}\left(-\frac{q^2}{4z}\right)\Big|_{\infty}^{d} = -\frac{1}{4\pi\epsilon_0}\frac{q^2}{4d}.$$

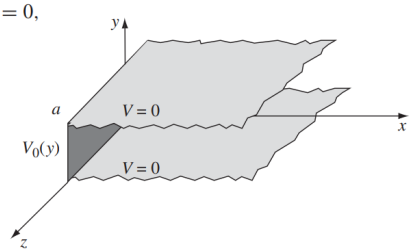
or we can also do this using eq 2.42

$$W = -\frac{1}{4\pi\epsilon_0}\frac{q^2}{2d}$$

and takings half of it:

$$W = -\frac{1}{4\pi\epsilon_0}\frac{q^2}{4d}$$

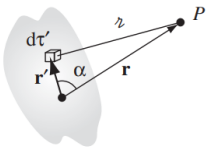
due to energy stored in the plate

3.3 separation of variables	
separation of variables	= look for a solution which is a product of functions, each depending on one of the coords > vb: potential V and charge density σ are specified and we search the potential of the interior
3.3.1 cartesian coordinates	
ex: grounded plates	we have two infinite grounded metal plates parallel to the xz plane, at x=0 and x=a > we have: $\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0,$ with boundary conditions (i) $V = 0$ when $y = 0$, (iii) $V = V_0(y)$ when $x = 0$, (ii) $V = 0$ when $y = a$, (iv) $V \to 0$ as $x \to \infty$.  we look at solutions of the form: $V(x, y) = X(x)Y(y).$ then we have: $\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} = 0.$ which is only possible if they're both constant: $\frac{1}{X}\frac{d^2X}{dx^2} = C_1 \quad \text{and} \quad \frac{1}{Y}\frac{d^2Y}{dy^2} = C_2, \quad \text{with} \quad C_1 + C_2 = 0.$ giving us solutions: $X(x) = Ae^{kx} + Be^{-kx}, \qquad Y(y) = C\sin ky + D\cos ky,$ thus the potential $V(x, y) = (Ae^{kx} + Be^{-kx})(C\sin ky + D\cos ky)$ now we look at the boundary conditions > (iv) requires A=0 > absorb B into C and D, giving us: $V(x, y) = e^{-kx}(C\sin ky + D\cos ky).$ (i) requires D=, while (ii) yields sinka=, from which we get: $V(x, y) = Ce^{-kx}\sin ky. \quad \text{with} \quad k = \frac{n\pi}{a}$ now we can take a linear combinations of potentials corresponding to each n $V = \alpha_1 V_1 + \alpha_2 V_2 + \alpha_3 V_3 + \ldots,$ they will still satisfy Laplace's equation: $\nabla^2 V = \alpha_1 \nabla^2 V_1 + \alpha_2 \nabla^2 V_2 + \ldots = 0\alpha_1 + 0\alpha_2 + \ldots = 0.$ thus we find a general solution: $V(x, y) = \sum_{n=1}^{\infty} C_n e^{-n\pi x/a}\sin(n\pi y/a).$ to determine C_n we use the (iii) boundary condition > this requires: $V(0, y) = \sum_{n=1}^{\infty} C_n \sin(n\pi y/a) = V_0(y).$ now multiply this by sin(n'πy/a) and integrate from 0 to a: $\sum_{n=1}^{\infty} C_n \int_0^a \sin(n\pi y/a)\,\sin(n'\pi y/a)\,dy = \int_0^a V_0(y)\sin(n'\pi y/a)\,dy.$ we have: $\int_0^a \sin(n\pi y/a)\,\sin(n'\pi y/a)\,dy = \begin{cases} 0, & \text{if } n' \neq n, \\ \frac{a}{2}, & \text{if } n' = n. \end{cases}$ thus all terms in the series drop out, exempt for n=n', thus we conclude: $C_n = \frac{2}{a}\int_0^a V_0(y)\sin(n\pi y/a)\,dy.$ which has the solution: $C_n = \frac{2V_0}{a}\int_0^a \sin(n\pi y/a)\,dy = \frac{2V_0}{n\pi}(1 - \cos n\pi) = \begin{cases} 0, & \text{if } n \text{ is even}, \\ \frac{4V_0}{n\pi}, & \text{if } n \text{ is odd}. \end{cases}$ giving us the expression: $V(x, y) = \frac{4V_0}{\pi}\sum_{n=1,3,5\ldots} \frac{1}{n}e^{-n\pi x/a}\sin(n\pi y/a).$

completeness of functions	a function is complete if any other function f(y) can be expressed as a lin combination of them $$f(y) = \sum_{n=1}^{\infty} C_n f_n(y).$$
orthogonal functions	a set of functions is orthogonal if: $\int_0^a f_n(y) f_{n'}(y)\, dy = 0 \qquad \text{for } n' \neq n.$
3.3.2 spherical coordinates	
potential V with azimuthal symmetry	Consider the Laplace equation in spherical coordinates: $$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial V}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial V}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 V}{\partial\phi^2} = 0.$$ now we have azimuthal symmetry, so the ϕ component doesn't change $$\frac{\partial}{\partial r}\left(r^2\frac{\partial V}{\partial r}\right) + \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial V}{\partial\theta}\right) = 0.$$ we separate the potential: $$V(r,\theta) = R(r)\Theta(\theta)$$ giving us the equation: $$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{1}{\Theta\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) = 0.$$ which each need to be constant: $$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) = l(l+1), \qquad \frac{1}{\Theta\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) = -l(l+1).$$ the first equation yields the solution: $$R(r) = Ar^l + \frac{B}{r^{l+1}}$$ whereas the second equation: $$\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) = -l(l+1)\sin\theta\,\Theta$$ the solutions are Legendre polynomials variable in cosθ $$\Theta(\theta) = P_l(\cos\theta).$$ which is most convenient in the Rodrigues formula: $$P_l(x) \equiv \frac{1}{2^l l!}\left(\frac{d}{dx}\right)^l (x^2-1)^l.$$ we thus get the formula: $$V(r,\theta) = \left(Ar^l + \frac{B}{r^{l+1}}\right) P_l(\cos\theta).$$ thus their superposition: $$V(r,\theta) = \sum_{l=0}^{\infty}\left(A_l r^l + \frac{B_l}{r^{l+1}}\right) P_l(\cos\theta).$$
3.4 multipole expansion	
3.4.1 approximate potentials at large distances	
electric dipole	consider two opposite charges +-q separated by a distance d > for r_- and r_+ the distance from -q and +q, we have a potential $$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\left(\frac{q}{r_+} - \frac{q}{r_-}\right)$$ and from the law of cosines, we find: $$r_\pm^2 = r^2 + (d/2)^2 \mp rd\cos\theta = r^2\left(1 \mp \frac{d}{r}\cos\theta + \frac{d^2}{4r^2}\right)$$ where we can approximate: $$\frac{1}{r_\pm} \cong \frac{1}{r}\left(1 \mp \frac{d}{r}\cos\theta\right)^{-1/2} \cong \frac{1}{r}\left(1 \pm \frac{d}{2r}\cos\theta\right)$$ thus: $$\frac{1}{r_+} - \frac{1}{r_-} \cong \frac{d}{r^2}\cos\theta,$$ giving us the potential: $$V(\mathbf{r}) \cong \frac{1}{4\pi\epsilon_0}\frac{qd\cos\theta}{r^2}$$

multipole expansion

consider any localized charge, like in the figure



> the potential at **r** is given by:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{1}{\mathcal{r}}\rho(\mathbf{r}')\,d\tau'.$$

using the law of cosines:

$$\mathcal{r}^2 = r^2 + (r')^2 - 2rr'\cos\alpha = r^2\left[1 + \left(\frac{r'}{r}\right)^2 - 2\left(\frac{r'}{r}\right)\cos\alpha\right]$$

where α is the angle between **r** and **r'**, thus:

$$\mathcal{r} = r\sqrt{1+\epsilon},$$

with:

$$\epsilon \equiv \left(\frac{r'}{r}\right)\left(\frac{r'}{r} - 2\cos\alpha\right)$$

for points outside the charge distribution, ε is much less then 1

> use the binomial expansion:

$$\frac{1}{\mathcal{r}} = \frac{1}{r}(1+\epsilon)^{-1/2} = \frac{1}{r}\left(1 - \frac{1}{2}\epsilon + \frac{3}{8}\epsilon^2 - \frac{5}{16}\epsilon^3 + \ldots\right)$$

or thus in terms of r,r' and α

$$\begin{aligned}\frac{1}{\mathcal{r}} = \frac{1}{r}\Bigg[&1 - \frac{1}{2}\left(\frac{r'}{r}\right)\left(\frac{r'}{r} - 2\cos\alpha\right) + \frac{3}{8}\left(\frac{r'}{r}\right)^2\left(\frac{r'}{r} - 2\cos\alpha\right)^2 \\ &- \frac{5}{16}\left(\frac{r'}{r}\right)^3\left(\frac{r'}{r} - 2\cos\alpha\right)^3 + \ldots\Bigg] \\ = \frac{1}{r}\Bigg[&1 + \left(\frac{r'}{r}\right)(\cos\alpha) + \left(\frac{r'}{r}\right)^2\left(\frac{3\cos^2\alpha - 1}{2}\right) \\ &+ \left(\frac{r'}{r}\right)^3\left(\frac{5\cos^3\alpha - 3\cos\alpha}{2}\right) + \ldots\Bigg].\end{aligned}$$

the terms in the parenthesis are Legendre polynomials

> we thus have:

$$\frac{1}{\mathcal{r}} = \frac{1}{r}\sum_{n=0}^{\infty}\left(\frac{r'}{r}\right)^n P_n(\cos\alpha).$$

we then have a potential:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\sum_{n=0}^{\infty}\frac{1}{r^{(n+1)}}\int (r')^n P_n(\cos\alpha)\rho(\mathbf{r}')\,d\tau',$$

>> for monopole n=1, dipole n=2, ...

3.4.2 the monopole and dipole terms

dipole moment

at large r, the multipole expansion is dominated by the monopole term:

$$V_{\text{mon}}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\frac{Q}{r}$$

however, if the total charge is 0, the dipole term is the dominant term:

$$V_{\text{dip}}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\frac{1}{r^2}\int r'\cos\alpha\,\rho(\mathbf{r}')\,d\tau'.$$

since α is the angle between **r'** and **r**, we have:

$$r'\cos\alpha = \hat{\mathbf{r}}\cdot\mathbf{r}',$$

thus the dipole potential can be written as:

$$V_{\text{dip}}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\frac{1}{r^2}\hat{\mathbf{r}}\cdot\int \mathbf{r}'\rho(\mathbf{r}')\,d\tau'.$$

now define the dipole moment of the distribution as:

$$\mathbf{p} \equiv \int \mathbf{r}'\rho(\mathbf{r}')\,d\tau',$$

such that: $V_{\text{dip}}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\frac{\mathbf{p}\cdot\hat{\mathbf{r}}}{r^2}$

>> for a collection of point charges we then have: $\mathbf{p} = \sum_{i=1}^{n} q_i\mathbf{r}_i'.$

3.4.3 origin of coordinates in multipole expansions	
origin and multipoles	consider a monopole at the origin > this is simply a point charge, with potential $(1/4\pi\varepsilon_0)q/r$ > however, now move the origin > the potential will also change: $(1/4\pi\epsilon_0)q/z.$ >> moving the origin changes the multipole expansion
3.4.4 the electric field of a dipole	
electric field of dipole	choose the coords such that **p** is at the origin and points in the z-direction > the potential at r,θ is: $$V_{\text{dip}}(r,\theta) = \frac{\hat{\mathbf{r}}\cdot\mathbf{p}}{4\pi\epsilon_0 r^2} = \frac{p\cos\theta}{4\pi\epsilon_0 r^2}.$$ take the negative gradient of V: $$E_r = -\frac{\partial V}{\partial r} = \frac{2p\cos\theta}{4\pi\epsilon_0 r^3},$$ $$E_\theta = -\frac{1}{r}\frac{\partial V}{\partial\theta} = \frac{p\sin\theta}{4\pi\epsilon_0 r^3}$$ $$E_\phi = -\frac{1}{r\sin\theta}\frac{\partial V}{\partial\phi} = 0.$$ then we find: $$\mathbf{E}_{\text{dip}}(r,\theta) = \frac{p}{4\pi\epsilon_0 r^3}(2\cos\theta\,\hat{\mathbf{r}} + \sin\theta\,\hat{\boldsymbol{\theta}}).$$ 